Jian Wei CHEONG

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Google Scholar

Curriculum Vitae

Professional Qualifications

2019 - 2023 **PhD, Physics**, *Nanyang Technological University (NTU)*, Singapore
2015 - 2019 **BSc, Physics, Honours**, *NTU*, Singapore
2010 - 2013 **Diploma, Electrical Engineering**, *Ngee Ann Polytechnic (NP)*, Singapore

Professional Experience

2024 - current **Postdoctoral Research Fellow**, *NTU*, Singapore
2023 **Project Officer**, *NTU*, Singapore
2013 - 2015 **Mandatory Military Service**, *Singapore Armed Forces*, Singapore
2012 **Intern**, *ST Electronics*, Singapore
2026 **Peer Reviewer**, *npj Quantum Information, Scientific Reports*,
2026/2025/2024 **Sub-reviewer**, *International Conference on Computional Science*, (ICCS)

Teaching Experience

2026/2025/2024 **Computational Nonlinear Lab (Physics Discovery Camp)**, *NTU*, Singapore
2025/2021 **PH3101 Quantum Mechanics 2**, *NTU*, Singapore
2020 **PH1199 Physics Lab 1B**, *NTU*, Singapore
2019 **PH1198 Physics Lab 1A**, *NTU*, Singapore
2024 - current **Student guidance (1 PhD)**, *NTU*, Singapore
current/2023 **Student guidance (3 MSc)**, *NTU*, Singapore

Awards & Achievements

2019 **Short-speech Contest Best Presentation** (PAP701 Graduate seminar module), NTU, Singapore
2017/2018 **Dean's List** (top 5% of cohort), NTU, Singapore
2016/2017 **NTU President Research Scholar** (completing URECA), NTU, Singapore
2011 **Director's List** (top 5% of cohort), NP, Singapore
2011 **Best Performance, Programmable Logic Device** (top student of cohort), NP, Singapore
2010 **Best Performance, Digital Electronics & Practice** (top student of cohort), NP, Singapore

Bibliography

Publications

1. Y. Ma, J. W. Cheong, K. Huang, H. Liu, X. Long, T. Xin, X. Nie, D. Lu, **Non-Markovianity supplies the magic behind indefinite-causal-order-based quantum refrigeration**, *Physical Review Letters (2026)* ***(Under Review)***.
2. Y. Wang, A. Pradana, J. W. Cheong, and L. Y. Chew, **Enhancing performance and operational regimes of the quantum Otto refrigerator with heat bath algorithmic cooling**, *Physical Review E, 113(5), 054112 (2026)*.
3. J. W. Cheong, A. Pradana, and L. Y. Chew, **Enhancing the performance of quantum switch refrigeration with an information pool of qubits**, *Physical Review Research, 8(1), 013272 (2026)*.
4. J. W. Cheong, A. Pradana, and L. Y. Chew, **Non-Markovian refrigeration and heat flow in the quantum switch**, *Physical Review A, 110(2), 022220 (2024)*.
5. L. Y. Chew, A. Pradana, L. He, and J. W. Cheong, **Stochastic thermodynamics of finite-tape information ratchet**, *European Physical Journal Special Topics (2023)*.
6. J. W. Cheong, A. Pradana, and L. Y. Chew, **Effects of non-Markovianity on daemonic ergotropy in the quantum switch**, *Physical Review A, 108(1), 012201 (2023)*.
7. L. He, J. W. Cheong, A. Pradana, and L. Y. Chew, **Effects of correlation in an information ratchet with finite tape**, *Physical Review E, 107(2), 024130 (2023)*.
8. J. W. Cheong, A. Pradana, and L. Y. Chew, **Communication advantage of quantum compositions of channels from non-Markovianity**, *Physical Review A, 106(5), 052410 (2022)*.
9. L. He, A. Pradana, J. W. Cheong, and L. Y. Chew, **Information processing second law for an information ratchet with finite tape**, *Physical Review E, 105(5), 054131 (2022)*.

Proceedings

1. L. Y. Chew, J. W. Cheong, and A. Pradana, **Thermodynamic Functionality of Non-detailed Balance Finite-Tape Information Ratchet**, *International Conference on Geometric Science of Information, Springer Nature Switzerland, pp. 173–181 (2025)*.

Talks

1. J. W. Cheong, A. Pradana, and L. Y. Chew, **Non-Markovian Quantum Refrigeration in the Quantum Switch**, *Quantum Thermodynamics Conference 2026 (QTD2026), Natal, Brazil, August 2026*.
2. J. W. Cheong, A. Pradana, and L. Y. Chew, **Maxwell's Uncertain Demon: Achieving Nonclassical and Enhanced Refrigeration when Causality Blurs**, *Asia-Pacific Conference on Networks and Complex Systems 2026 (APCNCS 2026), Singapore, June 2026*.

Posters

1. J. W. Cheong, A. Pradana, and L. Y. Chew[†], **Enhancement of quantum processes from indefinite causal order through non-Markovianity**, *29th International Conference on Statistical Physics (STATPHYS29), Florence, Italy, July 2025*.
2. J. W. Cheong[†], A. Pradana, and L. Y. Chew, **Non-Markovian refrigeration and heat flow in the quantum switch**, *Quantum Thermodynamics Conference 2025 (QTD2025), Singapore, July 2025*.
3. L. Y. Chew[†], J. W. Cheong, and A. Pradana, **Enhancement of quantum processes from indefinite causal order through non-Markovianity**, *XLV Dynamics Days Europe 2025 (DDE2025), Thessaloniki, Greece, June 2025*.
4. L. Y. Chew[†], L. He, A. Pradana, and J. W. Cheong, **Stochastic thermodynamics and correlation effects of finite-tape information ratchets**, *28th International Conference on Statistical Physics (STATPHYS28), Tokyo, Japan, August 2023*.

[†]Presenting author.

Technical Experience

	Skill	Level	Comment
Research	Non-Markovianity	■■■□	main research
	Indefinite causality	■■□□	main research
	Channel compositions	■■■□	main research
	Refrigeration	■■■□	application of main research
	Work engines	■■■□	application of main research
	Quantum communication	■■□□	application of main research
	Maxwell's demon	■■■□	main + secondary research
	Collision model	■■■□	main + secondary research
	Information ratchet	■■■□	secondary research
	Autonomous systems	■■□□	secondary research
	Quantum battery	■□□□	secondary research
	Entanglement monogamy	■■□□	side research
	Resource theory	■□□□	just starting (collaboration)
	Quantum reference frames	■□□□	just starting (collaboration)
	NMR	■□□□	light experience (collaboration)
	NV center magnetometers	■□□□	light experience
Programming	Python / Julia / R	■■■■	used in main work
	C / C++	■■□□	bachelor course, RCpp
	MATLAB	■■□□	bachelor course
	Haskell / Racket / Lisp	■□□□	personal projects
	Bash / POSIX sh	■■■□	personal Linux projects
	Quarto / LaTeX / Typst	■■■□	presentations, reports
	HTML / CSS	■■□□	personal website
Misc	Arduino & microcontrollers	■■□□	bachelor & diploma courses
	Blender	■□□□	scientific visualization
	LabVIEW	■□□□	bachelor & diploma courses

■□□□ basic knowledge ■■■□ extensive knowledge

■■□□ intermediate knowledge ■■■■ expert knowledge

Non-Quantum Project Experience

- **Strain estimation for hazard forecastings before and after 2011 Japan Tohoku earthquake** by analysis of seismic GPS displacement data and computation of seismic strains from velocity fields.
  ES7008 Geophysical Data Analysis, NTU
- **Variations in statistical complexity of genome sequences across species** by analysis of genome sequences from GenBank assembly with Baum-Welch algorithm.
  CE7412 Computational and Systems Biology, NTU
- **Detecting adversarial attack of deep neural networks for image recognition from image complexity** by Fast Gradient Sign Method (FGSM), DeepFool, One Pixel Attack, and Jacobian-Based Saliency Map Attack (JSMA).
  PH3502 Chaotic Dynamical Systems, NTU
- **Monte Carlo photon transport in multi-layered biological tissues** with Henyey-Greenstein scattering for diagnostic imaging and photon therapy.
  PH4505 Computational Physics, NTU
- **Monte Carlo simulation of periodic-driven Brownian particles** with Arrhenius equation, demonstrating dissipation-driven adaptation.
  PAP723 Numerical Methods for Physicists, NTU